

Faisal Mahmud Abir

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Professional Summary

Computer Science graduate focused on AI/ML engineering with strong research and project experience in computer vision, NLP, multimodal AI, and retrieval-augmented generation (RAG). Proficient in Python, PyTorch, Hugging Face, and Scikit-Learn for developing, evaluating, fine-tuning, and deploying machine learning models. Research experience includes vision transformers, self-supervised learning, knowledge distillation, and efficient deep learning.

Technical Skills

Programming & Data: Python, C, C++, JavaScript, SQL, NumPy, Pandas

AI/ML Frameworks & Techniques: PyTorch, Scikit-Learn, Hugging Face Transformers, RAG, Knowledge Distillation, Self-Supervised Learning, Model Fine-Tuning

Backend & Databases: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, ChromaDB

Cloud, Deployment & Tools: AWS, GCP, Docker, MLOps, Git, GitHub, Linux

Research Experience

Distilling-Hate

GitHub

Undergraduate Capstone Thesis • Manuscript Under Review at PLOS ONE

- Distilled a **336.7M-parameter BanglaBERT Large** teacher into a **13.7M-parameter student** using cross-entropy, KL divergence, and hidden-state MSE.
- Reduced inference computation by **96.8% in FLOPs**, improving suitability for resource-constrained and edge deployment.
- Achieved **0.5799 macro-F1** and a **2.5× improvement in multi-label F1** on the imbalanced BanglaMultiHate and BanTH test sets.

JSSL-GI: Self-Supervised ViT for GI Endoscopy Classification

GitHub

Undergraduate Research Project

- Designed a self-supervised pipeline combining **MAE reconstruction and multi-view InfoNCE** with an ImageNet-pretrained **ViT-Small** backbone.
- Applied **adaptive uncertainty-based loss weighting** to balance objectives and improve representation learning.
- Achieved **92.00% accuracy**, **67.30% macro-F1**, and **96.99% MCC** across **23 Hyper-Kvasir classes** while reducing pretraining from **400 to 30 epochs**.

Projects

MemeGuard AI

Live Demo — GitHub

Full-Stack Multimodal Hate Speech Moderation Platform

- Built a real-time **multimodal hate-speech detection system** ResNet-18 and DistilBERT.
- Deployed model inference through **Flask, REST APIs, and WebSockets** for real-time web and Telegram-based moderation.

MultiDoc AI

GitHub

RAG-Based Document Intelligence System

- Developed a **RAG-based document question-answering system** supporting PDF, DOCX, CSV, and XLSX inputs.
- Implemented semantic retrieval using **ChromaDB** with a **FastAPI** backend to provide context-aware responses.

MegaCloud Storage Management

GitHub

Distributed Cloud Storage Management System

- Built a unified cloud storage management system integrating **Google Drive and Dropbox** for cross-platform file operations.
- Implemented **chunk-based streaming for large-file transfers**, enabling efficient handling of files without loading entire files into memory.

Industry Experience

Codveda Technologies

September 2026 – Present

Data Analysis Intern

Remote

- Cleaned, transformed, and analyzed datasets using **Python, Pandas, and NumPy**, performing exploratory data analysis (EDA) to identify patterns, trends, correlations, and data-quality issues.
- Created statistical visualizations and analytical reports using **Matplotlib and Seaborn** to communicate insights and support data-driven interpretation.
- Developed and evaluated **machine learning models** using **Scikit-Learn**, applying data preprocessing, feature engineering, model comparison, and performance evaluation.

Education

North South University

Spring 2022 – Spring 2026

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

Leadership Experience

IEEE NSU Industry Applications Society (IAS) Student Branch Chapter

2023 – 2025

Lead Coordinator, Events & Management Team

Dhaka, Bangladesh

- Led **24+ technical and professional development programs**, coordinating logistics, budgets, speakers, sponsors, and faculty.
- Received the **Best Lead Award (2024–2025)** for leadership and team coordination.